Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

V SEMESTER B.TECH.
END SEMESTER EXAMINATION, NOVEMBER - DECEMBER 2024
ELE 3127 - FOUNDATIONS EV AND HYBRID VEHICLES

Time: 3 Hours **Date: 30 NOVEMBER 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A.	Examine the primary factors contributing to the growth of the Electric Vehicle (EV) market in India.	***4***
1B.	Summarize the key players in the Indian EV market.	***2***
1C.	Classify the specific types of electric vehicles that these major players focus on.	***4***
2A.	Describe the challenges arise from the high voltage present in Electric Vehicle (EV) systems?	***3***
2B.	Categorize the additional safety measures necessary to manage these high-voltage systems effectively.	***2***
2C.	Analyze how the cooling system in Electric Vehicles (EVs) differs from that in Internal Combustion Engine (ICE) vehicles?	***5***
3A.	Illustrate the primary energy sources used in Hybrid Electric Vehicles (HEVs) and how they interact.	***4***
3B.	Distinguish how Plug-in Hybrid Electric Vehicles (PHEVs) differ from conventional Hybrid Electric Vehicles (HEVs).	***3***
3C.	Evaluate the benefits that PHEVs provide in terms of electric-only driving range compared to traditional HEVs?	***3***
4A.	Solve for the current being used to charge a 10Ah battery at a 0.5C rate, and find out how long it will take to fully charge?	***2***
4B.	Categorize the environmental impacts related to battery production and disposal?	***4***
4C.	Assess how industries are addressing the environmental concerns related to battery production and disposal?	***4***
5A.	Explain the benefits of utilizing Permanent Magnet Synchronous Motors (PMSMs) in electric vehicles?	***4***
5B.	Examine how Brushless DC (BLDC) motors and PMSMs differ in terms of torque generation?	***4***
5C.	Distinguish the primary applications of Brushless DC (BLDC) motors in electric vehicles in relation to those of PMSMs?	***2***